Standardisation of materials and improving production processes are necessary to increase efficiency. For instance, some 3D printers made by using fused deposition modelling (FDM) technique can be easily recycled once they can no longer be used.

People have to simply ensure that they do not dump their used products with other household wastes, and only certified recyclers or dismantlers handle them. While donating or selling them to others, they should check that the products are in working condition. By preferring recycled materials, more jobs can be created in this sector. According to 'Global E-waste Management Market Size, Market Share, Application Analysis, Regional Outlook, Growth Trends, Key Players, Competitive Strategies and Forecasts, 2018 To 2026' report, global e-waste management market is expected to grow at a CAGR of 5.7 per cent during the forecast period from 2018 to 2026. Extra Carbon and Binbag are among the numerous companies that have already come up in this sector to aid people in India.

In formal processing, waste segregation is done at first to recover valuable materials beforehand using techniques like electrolysis, filtration, centrifugation, etc. Goods that can be reused are repaired and resold at a lower price. This way fewer products need to be manufactured annually, leading to lower greenhouse gas emissions.

Hazardous materials are separated and treated to reduce their harmfulness. If materials are recyclable, they can be easily converted into something useful again, thus decreasing waste stream concentration. Workshop Q, a manufacturing company in Jaipur, uses waste having eco-friendly materials to make household items.

Ministry of Environment, Forest and Climate Change, government of India, has notified E-Waste (Management) Rules in 2016 and E-Waste (Management) Amendment Rules in 2018 for eco-friendly recycling and reducing e-waste. Following this, companies like GreyOrange have taken up the responsibility to recycle and dispose of their products. Though a lot of initiatives are being taken, further adoption of strict regulations and more programmes to create awareness among the consumers is the need of the hour.

—Ayushee Sharma

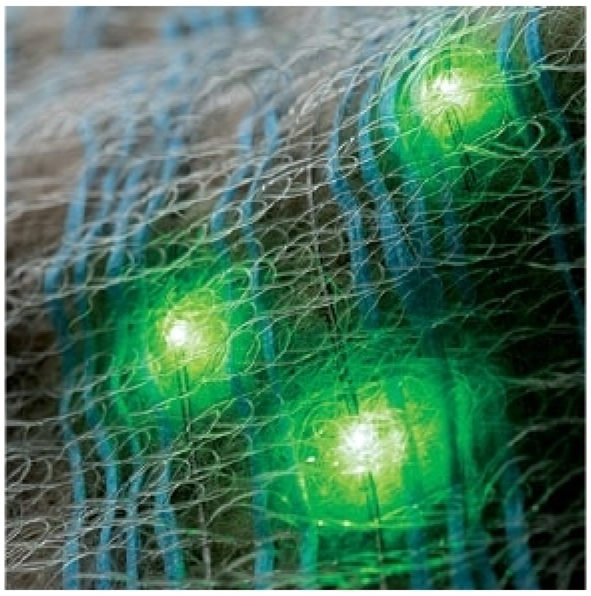
2 SMART FABRICS CAN HELP BOOST SMART WEARABLE INDUSTRY

Smart fabrics were initially developed for performance-enhancing purpose for professional athletes, but now their applications are expanding across various industries

The traditional concept of clothing is steadily witnessing a change with advances in technological innovations. Smart fabrics, also known e-textiles or intelligent fabrics, are making new waves in the smart wearable industry.

As the name suggests, smart fabrics can deliver many smart functions unlike conventional fabrics. These smart fabrics are developed by integrating technologies into fabrics that can interact, sense and react to environmental stimuli. These have the ability of performing several electronic functions, from heating up fabrics to collecting data and sending alerts. Smart fabrics are currently in their infancy; however, these are set to gain a firm upper hand in the clothing industry in the future.

In smart fabrics, digital components are woven into traditional fabrics. These components can be biomedical sensors, batteries, microcontrollers, fibre-optics, electronic chips, wearable antennae and so on. These fabrics are active and vibrant materials that possess actuation and sensing properties. Components are incorporated using various methods, such as conductive fibres and multilayer 3D printing.



Lighting up: a woven fabric that incorporates fibre-based LEDs (Credit: https://physicsworld.com)

Many people think that fabrics with embedded components are not comfortable to wear and can restrict body movement. On the contrary, these fabrics are flexible, light, hassle-free and user-friendly. Their main motive is to give a sense of safety and added value to the wearer. These can regulate body temperature and change colours as per body heat. These garments are very much like ordinary clothes except that these can sense and act to environmental radiation and hazards in a predetermined way.

Smart textiles can be categorised into three types: passive, active and ultra-smart, as explained next.

Passive smart fabrics. Wearable clothing in this category is less complex and novel in terms of functionality. These have features that can sense and gauge movements of users and the surrounding environment. These fabrics are anti-microbial, anti-static, anti-odour, bullet-proof, etc.

Active smart fabrics. Both sensors and actuators are used in such smart textiles. These are aptly called active smart fabrics because these can easily switch their functionalities as per (unstable) environments. Active smart textiles are capable of heat storage and regulation. Besides, these are water-resistant and vapour-permeable (hydrophilic/non-porous).

Ultra-smart fabrics. These tex-